

ORIGINAL ARTICLE

Incidence, Timing and Clinical Significance of Immune-Mediated Myocardial Injury and Myocarditis After Gene Replacement Therapy: A Systematic Review and Meta-Analysis

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BACKGROUND: Although adeno-associated virus (AAV) gene-replacement therapy is a potentially transformative therapy for severe genetic diseases, its cardiac immunotoxicity may challenge broad clinical use.

METHODS: Medical Literature Analysis and Retrieval System Online, Embase, and PubMed databases were searched from January 2005 to March 2025. Studies including patients treated with AAV-replacement therapy were deemed eligible. Prespecified items (type of vector, dose, timing, and clinical significance of the adverse event) were extracted by 2 independent observers. Random-effects models were fitted using restricted maximum likelihood estimation and the method of Hartung, Knapp, Sidik, and Jonkman (International Prospective Register of Systematic Reviews [PROSPERO], REGISTRATION: URL: <https://www.crd.york.ac.uk/PROSPERO/>; Unique identifier: CRD420251046546). VigiBase and the US Food and Drug Administration Adverse Event Reporting System were searched for the occurrence of myocarditis with commercially available AAV-replacement drugs.

RESULTS: Eighty studies including 1939 human patients were analyzed. A total of 734 adverse events were reported over 2122 patient-years of pooled observation. Seventy-one cases of myocarditis were identified. The pooled incidence rate per 100 patient-years was 8.6 (95% CI, 5.8–10.7; $P=63.2\%$). Events occurred in patients with Duchenne muscular dystrophy, spinal muscular atrophy, and X-linked myotubular myopathy, with recombinant AAVs and adeno-associated virus serotype 8. All received an intravenous dose $>10^{13}$ vector genomes per kilogram body weight. Myocardial injury peaked in week 1 after injection (90% [95% CI, 85.7%–96.2%]; $P=43.2\%$), whereas myocarditis occurred mostly after week 2 (55% [95% CI, 48.7%–65.2%]; $P=33.1\%$), with no cases after the first month. Most myocarditis/myocardial injury did not have a relevant clinical impact (62, 87%), with only 8 (12%) cases having transient left ventricular dysfunction. The latter recovered during the follow-up. The only death occurred in the setting of cytokine-mediated capillary leak syndrome with cardiac dysfunction. Myocarditis occurred in relation to delandistrogene moxeparvovec (1/16 [6%]) and onasemnogene abeparvovec (14/217 [6%]) in the Food and Drug Administration Adverse Event Reporting System and VigiBase.

CONCLUSIONS: Immune-mediated myocarditis/myocardial injury after systemic AAV gene therapy occurred in <10 per 100 patient-years (peaking at weeks 1–3) and was usually clinically mild, with a minority showing transient dysfunction. All events followed intravenous doses $>1 \times 10^{13}$ vector genomes per kilogram body weight, clustered in neuromuscular programs, and were associated with certain recombinant capsids/serotypes. These data support intensive early cardiac monitoring, cautious dose selection, and delivery strategies to minimize systemic exposure.

Key Words: capsid ■ humans ■ incidence ■ myocarditis ■ serogroup

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WHAT IS NEW?

- We pooled 80 trials and cohorts (1939 patients; 2122 patient-years) of systemic adeno-associated virus gene-replacement therapy to define the true incidence, timing and severity of immune-mediated myocardial injury and myocarditis.
- Myocarditis/myocardial injury occurred in about 6% of treated patients (8.6 events/100 patient-years), clustered in weeks 1 to 3 (peak at week 2), and was almost never seen after the first month. Events arose only after intravenous doses >1×10¹³ vg/kg, mainly in neuromuscular programs, and were usually mild and reversible.
- These data support intensive early cardiac monitoring, cautious dose selection and delivery strategies to minimize systemic exposure.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Clinicians involved in adeno-associated virus gene-therapy programs should expect that most immune-mediated myocardial events will occur in the first month, especially weeks 1 to 3, and implement protocolized troponin/ECG checks plus rapid access to echocardiography/cardiovascular magnetic resonance in this window.
- Because almost all myocarditis episodes were mild and reversible with steroids or other immunosuppression, early recognition can avoid unnecessary alarm while enabling timely escalation in the small subset who develop ventricular dysfunction or systemic inflammation.
- Dose, route of delivery, indication and adeno-associated virus platform shape risk; programs should minimize systemic exposure in high-risk neuromuscular indications and embed standardized cardiac monitoring and adjudication to guide safer dosing and targeted immunosuppression.

Nonstandard Abbreviations and Acronyms

AAV	adeno-associated virus
AE	adverse event
cTn	cardiac troponin
DMD	Duchenne muscular dystrophy
ECG	electrocardiogram
FAERS	Food and Drug Administration Adverse Event Reporting System
TMA	thrombotic microangiopathy
vg/kg	vector genomes per kilogram body weight

Adeno-associated virus (AAV)-associated gene replacement therapies hold great therapeutic promise, but complications, in particular myocarditis and other forms of myocardial injury, have emerged

as a potential barrier to broader clinical use.^{1–4} Early postinfusion cardiac events reported across trials range from isolated troponin elevation to clinically significant myocardial dysfunction and arrhythmia, with proposed mechanisms spanning from early innate immune/complement-mediated endothelial injury to later cell-mediated inflammatory myocarditis.⁵ Because many gene therapy programs enroll small cohorts and report heterogeneous end points, the true incidence, timing, severity, and clinical consequences of AAV-associated myocarditis remain poorly defined.^{6,7} Moreover, reliance on single biomarkers (eg, troponin) without contemporary imaging or histology conflates disparate pathological processes and limits mechanistic insight and management guidance.^{5,8} To address this gap, we performed a systematic review and meta-analysis of published studies and pharmacovigilance reports of AAV gene therapy to quantify the incidence and time course of myocardial injury after AAV administration, characterize clinical phenotypes and outcomes, and identify potential vector- and patient-level factors associated with increased risk. These data will inform surveillance and mitigation strategies for future trials.



METHODS

This meta-analysis is reported in line with the Meta-analyses of Observational Studies in Epidemiology⁹ guidance and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (Table S1)¹⁰ statements and was registered on the International Prospective Register of Systematic Reviews (PROSPERO; CRD420251046546).¹¹ Data and analytical methods will be available upon reasonable request, and no institutional review board approval was needed for this study.

Search Strategy

The electronic databases Medical Literature Analysis and Retrieval System Online, Embase, and PubMed were comprehensively searched for English-language papers from January 2005 through March 2025, as previously reported.¹² The search syntax was designed for prospective and retrospective studies including patients treated with AAV gene-replacement therapy and is detailed in Table S2. Ongoing trial registry records were excluded. Any discrepancies in study selection were resolved through discussion and consensus between the 2 reviewers (N.M., E.S., Figure S1). A third reviewer (K.H.) was available for arbitration. If key data were missing from the published report, we planned to contact the corresponding authors for clarification. If these data remained unavailable after inquiry/no contact was possible, the study was included in analyses for which it provided sufficient data.

Data Extraction and Quality Assessment

Full text-review and prespecified items extraction such as number of patients, disease treated, observation time, type of vector used, dose, immunosuppressive regimen, and type and

clinical significance of the adverse event (AE) were performed independently at the study level by 2 independent reviewers (N.M., E.S.). Quality assessment was performed using the Cochrane Risk of Bias tool for clinical trial and observational studies.^{13,14}

Outcomes

The primary outcome of interest was to assess the incidence of AAV gene-replacement therapy immune-mediated AEs, myocardial injury, and myocarditis at any time during the published follow-up. Specifically, acute myocarditis was defined by the presence of cardiac symptoms (eg, chest pain, dyspnea, palpitations, syncope), an elevated cardiac troponin (cTn) above the 99th percentile, and abnormal electrocardiographic, echocardiographic, cardiovascular magnetic resonance, or histopathologic findings on biopsy or postmortem evaluation in the absence of flow-limiting coronary artery disease.^{15,16} Myocardial injury was defined as a condition defined by a cTn level above the 99th percentile upper reference limit.¹⁵ In patients with an increased baseline level of high-sensitivity troponin, an increase of at least 1.5 from baseline value was considered as significant.¹⁶ AEs were assessed for clinical significance based on the need for hospitalization or end-organ functional damage. Resolution was determined if the complication had resolved by the end of the study period.

Real-World Database Search

We performed an observational cross-sectional study focusing on the reporting of myocarditis, hepatitis, and thrombotic microangiopathy (TMA) using 2 international pharmacovigilance databases, World Health Organization global database of individual case safety reports¹⁷ and the Food and Drug Administration AE Reporting System (FAERS).¹⁸ Commercially available gene-replacement therapy drugs were searched, and 2 independent investigators (N.M. and E.S.) adjudicated the potential occurrence of study-defined myocarditis for each drug based on the information provided in the database reports (eg, Medical Dictionary for Regulatory Activities terms, narrative summaries if available). The adjudication aimed to align pharmacovigilance reports with the clinical definitions used for the analysis. Disagreements in adjudication were resolved as detailed above.

Statistical Analysis

A random-effects meta-analysis model using restricted maximum likelihood estimation (REML) was fitted, and the method of Hartung, Knapp, Sidik, and Jonkman adjustments was used to synthesize estimates and CIs.^{19,20} For individual study proportions we computed exact (Clopper-Pearson) 95% confidence limits. For pooled proportions we fitted random-effects models on the logit scale, applied the Hartung-Knapp-Sidik-Jonkman adjustment, and back-transformed pooled estimates and confidence bounds to the proportion scale via the inverse-logit function. Because both exact binomial intervals and the logit/inverse-logit transformation are nonlinear on the original proportion scale, study-level and pooled CIs may appear asymmetrical (this effect is magnified for proportions near 0% or 100% and in small studies). For incidence rates per 100 patient-years,

we computed Poisson/log-transformed estimates and back-transformed to the rate scale.

Heterogeneity was quantified using the I^2 statistic, and P values were calculated from a χ^2 test.²¹ Strict thresholds for interpretation are not recommended, but in general, an I^2 statistic at or above 50% and a χ^2 test $P < 0.10$ may be considered representative of substantial or considerable heterogeneity.²¹ Assessment of small-study effects, encompassing publication bias, outcome reporting bias, and clinical heterogeneity, was conducted statistically using the Egger test. To formally test whether AEs differed between clinical trials and observational studies, we performed a random-effects meta-regression on the logit-transformed incidence proportions. Person-years were extracted/estimated from trials (directly when reported, or derived from patient count $\times$ median/mean follow-up). All analyses were performed in the statistical programming environment R, version 4.3.1 (RStudio).

RESULTS

A total of 756 abstracts were initially screened; 182 full-text articles were then assessed for eligibility, yielding 80 studies that together enrolled 1939 patients^{22–101} (Figure S1 and Table S3). Full details on each included report—disease treated, vector platform, administered dose, and immunosuppressive strategy are provided in Table 1. Most studies were interventional clinical trials (61/80, 76%), with the remainder being observational cohorts (19/80, 24%). Seventy studies (88%) enrolled fewer than 50 participants, and vector doses spanned from 3.3×10^8 vector genomes per kilogram body weight (vg/kg) up to 1.1×10^{14} vg/kg. Immunomodulatory approaches clustered into 3 categories: reactive corticosteroid use after suspected AEs (reported in 28 studies, 35%, covering 915 patients); routine peri-infusion corticosteroid protocols (42 studies, 53%, 968 patients); and combined regimens incorporating peri-infusion corticosteroids plus mTOR inhibitors, calcineurin inhibitors, and anti-CD20 monoclonal antibodies (9 studies, 12%, 56 patients). One study did not specify the immunosuppression regimen.²²

Risk-of-bias appraisal findings are summarized in Figures S2 through S4. Among the interventional trials, 4 (5%) were judged at moderate risk with respect to outcome measurement, and 4 (5%) had unclear risk for selective reporting, yielding an overall moderate-bias prevalence of 3% across trials. For observational studies, 4 (17%) raised concerns related to participant selection and a 4 (17%) about uncontrolled confounding, with the pooled overall risk of bias estimated at 8%. Missing data details for each included report are provided in Table S4. A random-effects meta-regression on logit-transformed incidence proportions formally tested whether AE rates differed between study designs and found no statistically significant difference ($\beta = -0.580$; odds ratio, 0.51 [95% CI, 0.24–1.12]; $P = 0.081$; Table S5).

Table 1. Overview of the Studies Included by Target Disease

Disease treated	Number of studies	Sample size			Vector Used	Administration route	Doses	Immunosuppressive regimen			Myocardial adverse events
		n<10	10–50	n>50				Reactive Corticosteroids	Pre + post-treatment corticosteroids	Pre + post-treatment corticosteroids + mTOR inhib and calcineurin inhib and anti-cd-20	
Danon ²²	1	1	0	0	AAV-9	Systemic i.v.	6.7×10^{13} vg/kg– 1.1×10^{14} vg/kg	0	0	1	0
Anderson-Fabry ²³	1	1	0	0	AAV-2	Systemic i.v.	1×10^{13} vg/kg – 5×10^{13} vg/kg	0	1	0	0
Duchenne Muscular Dystrophy ^{24–34}	11	8	2	1	AAV-8, AAV-9, rAAVrh74, rAAV-2/8	Systemic i.v.	1×10^{12} vg/kg – 3×10^{13} vg/kg	1	6	3	4
Limb-girdle Muscular Dystrophy 2B ^{35,36}	2	2	0	0	rAAV-1, rAAVrh74	Systemic i.v.	1×10^{11} vg/kg– 7.4×10^{13} vg/kg	1	1	0	0
Pompe ^{37–39}	3	3	0	0	rAAV, AAV-8, AAV-1	Systemic i.v. Intradiaphragmatic	1×10^{12} vg/kg– 5×10^{12} vg/kg	1	1	1	0
Spinal Muscular Atrophy ^{40–63}	24	8	13	3	scAAV9-FL-SMNC-DNA	Systemic i.v.	6.7×10^{13} vg/Kg– 1.1×10^{14} vg/kg	0	24	0	65
X-Linked Myotubular Myopathy ⁶⁴	1	0	1	0	AAV-8	Systemic i.v.	1.3×10^{14} vg/Kg– 3.5×10^{14} vg/kg	0	1	0	2
Hemophilia A ^{65–71}	7	1	4	2	AAV-3, AAV-5, rAAV-6	Systemic i.v.	9×10^{11} vg/kg– 6×10^{13} vg/kg	5	2	0	0
Becker Muscular Dystrophy ⁷²	1	1	0	0	AAV-1	Intramuscular	3×10^{11} vg/kg– 6×10^{13} vg/kg	0	1	0	0
Chronic Heart Failure ^{88–92}	5	2	2	1	AAV-1, Ad5-FDG	Coronary	1.4×10^{11} vg/kg– 1×10^{13} vg/kg	5	0	0	0
Stable Angina Pectoris ^{93–95}	3	0	3	0	Ad5-FDG	Coronary	3.3×10^9 vg/kg – 1×10^{11} vg/kg	3	0	0	0
Hemophilia B ^{72–78}	7	2	3	2	rAAV-2, AAV-5, AAV-2/8, AAV-2, AAVs3	Systemic i.v.	8×10^{10} vg/kg– 2×10^{13} vg/kg	5	2	0	0
AADC Deficiency ^{78,79}	1	0	1	0	AAV-2	Intracerebral	1.8×10^{11} vg/kg– 2.4×10^{13} vg/kg	1	0	0	0
Autosomal Recessive Deafness ⁸⁰	1	1	0	0	AAV-1	Binaural	9×10^{11} vg/kg– 1.5×10^{12} vg/kg	1	0	0	0
Fronto Temporal Dementia ⁸¹	1	0	1	0	AAV-9	Intracerebral	2.1×10^{13} vg/kg– 4.2×10^{13} vg/kg	0	0	1	0
Crigler-Najar ⁸²	1	1	0	0	AAV-8	Systemic i.v.	2×10^{12} vg/kg– 5×10^{12} vg/kg	0	0	1	0
Mucopolysaccharidosis Type IIIA/IIIB/IV ^{83–86}	4	4	0	0	AAV-2/5, AAV-2/8, rAAV	Intracerebral	6×10^{11} vg/kg– 6×10^{12} vg/kg	0	1	3	0
Acute Intermittent Porphyria ⁸⁶	1	1	0	0	rAAV-2/5	Systemic i.v.	5×10^{11} vg/kg– 1.8×10^{13} vg/kg	1	0	0	0
Lipoprotein Lipase Deficiency ⁸⁷	1	1	0	0	AAV-1	Intramuscular	1×10^{12} vg/kg– 5×10^{12} vg/kg	0	1	0	0

(Continued)

Table 1. Continued

Disease treated	Number of studies	Sample size			Vector Used	Administration route	Doses	Immunosuppressive regimen			Myocardial adverse events
		n<10	10–50	n>50				Reactive Corticosteroids	Pre + post-treatment corticosteroids	Pre + post-treatment corticosteroids + mTOR inhib and calcineurin inhib and anti-cd-20	
Glycogen Storage Disease Type Ia ⁹⁸	1	0	1	0	AAV-2	Systemic i.v.	2 × 10 ¹² vg/kg–6 × 10 ¹³ vg/kg	1	1	0	0
Tay-Sachs ⁹⁹	1	1	0	0	rAAVrh8	Intacerebral	1 × 10 ¹³ vg/kg	0	1	0	0
HIV ¹⁰⁰	1	0	1	0	AVV-1	Intramuscular	1 × 10 ¹² vg/kg–1 × 10 ¹⁴ vg/kg	1	0	0	0
AAT Deficiency ¹⁰¹	1	0	1	0	rAAV, AAV-2	Intramuscular	0.5 × 10 ¹³ vg/kg–5 × 10 ¹³ vg/kg	0	1	0	0

AAV, adenovirus ; AADC Def, aromatic L-amino acid decarboxylase deficiency; AAT, Alfa-1 antitrypsin.

Pooled Incidence of Immune-Mediated AEs, Myocardial Injury, and Myocarditis

A total of 734 AEs were reported over 2122 patient-years of pooled observation. Hepatotoxicity occurred in 653 patients, followed by myocardial injury in 71 and TMA in 10. Six treatment-related deaths have been reported (Table S6). A total of 31 (39%) studies did not report any AE, whereas 9 (11%) presented an AE in >80% of the patients included (Figure S5).

Pooled incidences were, respectively: 30% (95% CI, 22.5%–38.9%; $P=83.1\%$, $P<0.01$) for all AEs; 6.2% (95% CI, 4.6%–8.1%; $P=46\%$, $P=0.07$) for the occurrence of myocardial injury/myocarditis; and 4.7% (95% CI, 3%–5.3%; $P=46.1\%$, $P=0.809$) for treatment-related death (Figure 1). Pooled incidence rate per 100 patient-years were: 33.8 (95% CI, 23.4%–43.1%; $P=46.1\%$, $P<0.01$) for all AEs and 8.6 (95% CI, 5.8%–10.7%; $P=63.2\%$, $P=0.13$) for myocardial injury/myocarditis (Figure 1).

When the meta-analysis was restricted to confirmed myocarditis cases only (those with imaging and cardiovascular magnetic resonance/histopathology confirmation), the pooled incidence and pooled incidence rate decreased to 2.9% (95% CI, 1.8%–3.7%; $P=58.4\%$, $P=0.21$) and to 3.6 (95% CI, 2.4%–4.7; $P=55.3\%$, $P=0.34$), respectively (Figure 1).

Timing and Clinical Significance of Myocardial Injury and Myocarditis

Of the 71 cases with reported myocardial injury, 40 had a diagnosis of myocarditis, defined by the presence at the time of diagnosis of wall motion abnormalities or decreased left ventricular ejection fraction on echocardiogram. Cardiac magnetic resonance was performed in 2 patients, and no myocardial biopsy was ever reported.

For the remaining 31 cases, no report of any complementary cardiac investigation or description of the clinical picture was reported.^{40,50,57,59}

Two different clinical and temporal patterns were observed. Myocardial injury peaked in the first week after injection (90% [95% CI, 85.7%–96.2%; $P=43.2\%$]), up to week 2 (7% [95% CI, 3.2%–14.2%; $P=58.8\%$]). Myocarditis mostly occurred in week 2 (55% [95% CI, 48.7%–65.2%; $P=33.1\%$]) and week 3 (38% [95% CI, 31.9%–43.6%; $P=54.2\%$]). No cases of were reported after the first month postinjection (Figure 2).

Detailed descriptions of each reported case of myocardial injury/myocarditis after gene-replacement therapy are presented in Table 2. Thirty-six (51%) patients presented with concomitant hepatotoxicity,^{25,27,31,34,41,48–51,58,60,64} and all but 1 case resolved by the end of the study period (Figure 2).^{25,27,34} All patients received preinfusion oral or intravenous steroids at a dose of 1 mg/kg and continued the treatment postinfusion. Cases 4, 70, and 71 (Table 2) were treated after the diagnosis of myocarditis with eculizumab, tocilizumab, anakinra, sirolimus, and mycophenolate mofetil, respectively.

Two pediatric Duchenne muscular dystrophy (DMD) recipients (Cases 1 and 2 of Table 2), both treated with high systemic doses of microdystrophin vectors (AAV serotype rh74-MHCK7 and AAV serotype 8 at 1.3 × 10¹⁴ vg/kg), presented at ≈3 weeks featured diffuse muscle weakness with new cardiac involvement (troponin rise, focal wall-motion abnormalities, cardiovascular magnetic resonance T2 signal consistent with edema and reduced left ventricular ejection fraction); the investigators interpreted these syndromes as transgene-directed immune reactions against sequence elements present in the therapeutic microdystrophin but absent in the patients' native dystrophin,

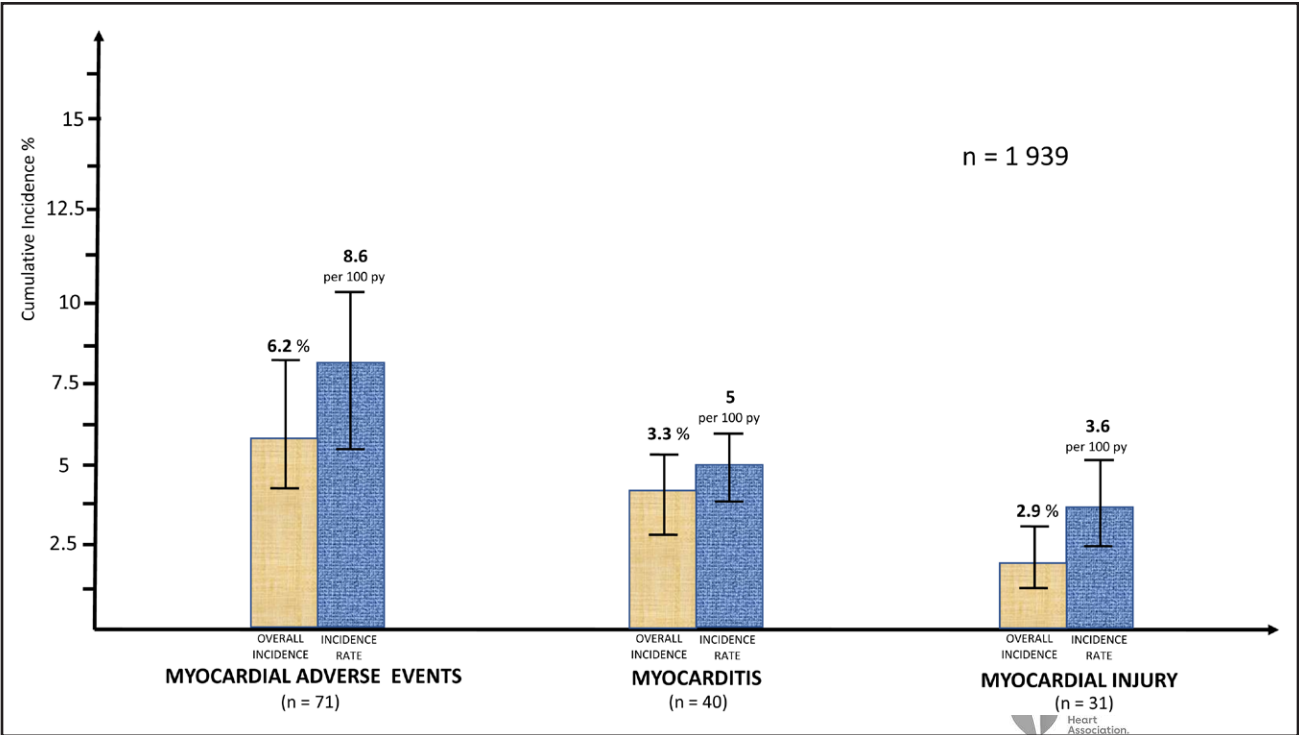


Figure 1. Pooled and annual incidence of myocardial adverse events after adeno-associated virus (AAV) gene replacement therapy. Pooled incidence of myocardial adverse events cases are presented with their 95% CIs. py indicates patient-years.

and both episodes resolved with corticosteroid-based immunosuppression (Table 2).^{25,27} By contrast, the Case 3 (Table 2) followed a different time course: within hours of infusion the patient developed fever, vomiting, early troponin elevation and transient hypotension with no echocardiographic change; the report attributed this immediate systemic syndrome more plausibly to an early capsid/innate immune response

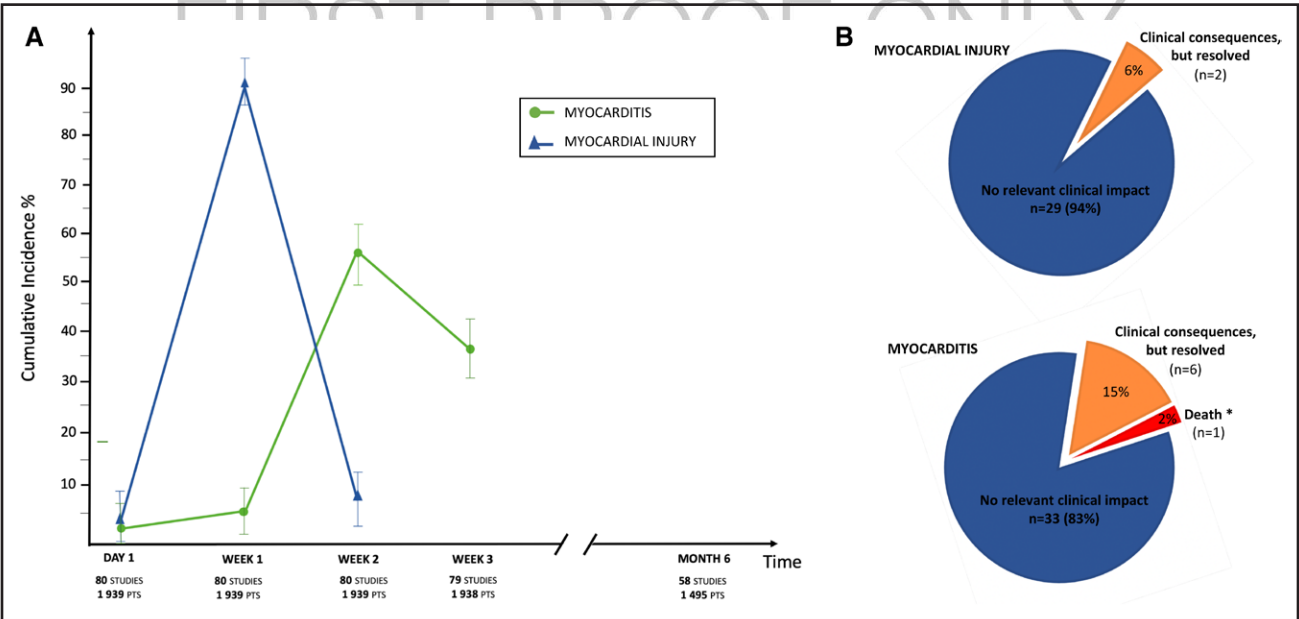


Figure 2. Timing of onset and clinical impact of gene replacement related myocarditis and myocardial injury. In **A**, the timing of onset of myocarditis and myocardial injury is presented. In **B**, the clinical impact of the myocarditis episodes and myocardial injuries are described. * = episode of death occurred in the context of a Cytokine-mediated Capillary Leak Syndrome with an associated cardiac dysfunction due to treatment acute toxic effect on a background of a preexisting cardiomyopathy. Concomitant myocarditis was possible, but did not probably represented the direct cause of death.

Table 2. Characteristics of the 71 Myocarditis Episodes Related to Immune Activation After Gene Replacement Therapy

Pa-tients	Study	Age	Dis-ease	AAV delivery vector	Dose	Prede-livery immu-nosup-pression	Post-therapy immu-no-suppres-sion	Preex-isting condi-tions	Clinical Course	Echocar-diogram	Explanation	Out-come
Case 1	Zaidman et al 2023 ²⁷	7	DMD	AAVrh74 + MHCK7 promoter	1.3×10^{14} vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o. and I.V.)	Large deletion from exon 8–21 in dystrophin gene	3 wk: severe diffuse muscular weakness. Myositis. ↑ TnnI, focal wall motion abnormalities decreased EF, ↑ T2 at Cardiac MR (CMR)	focal wall motion abnormalities decreased EF	Microdystrophin construct with hing 1 and the beginning of the spectrin like domain, absent in the patients and acting as "nonself" epitope	Re-solved during follow-up
Case 2	Bonneman et al 2023 ²⁵	9	DMD	AAV-8	1.3×10^{14} vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o. and I.V.)	Large deletion from exon 8–21 in dystrophin gene	3 wk: severe diffuse muscular weakness. Myositis. ↑ TnnI, focal wall motion abnormalities decreased EF, ↑ T2 at Cardiac MR (CMR)	focal wall motion abnormalities decreased EF	Microdystrophin construct with hing 1 and the beginning of the spectrin like domain, absent in the patients and acting as non-self epitope	Re-solved during follow-up
Case 3	Mendell et al 2025 ³²	...	DMD	rAAVrh74	1.3×10^{14} vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Day 0: ↑ Tnn, vomiting, fever. Transient spontaneous hypotension. No significant change in the echocardiogram.	Normal	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 4	Lek et al 2023 ³¹	27	DMD	rAAV-9 dSaCas9	1×10^{14} vg/kg	Cortico-steroids (p.o.)	Day 0: Cortico-steroids Day 1: Eculizumab Day 6: Eculizumab, Tocilizumab, Anakinra	Lean muscle mass 45% Restrictive Pul-monary Defect Mild DMD Cardio-myopa-thy (EF 55%)	Day 1: PVCs, ↓ PLT Day 5: ↓ EF (45–50%), ↑ TnnI, pericar-dial effusion. Day 6: ARDS; Day 8: ECMO and death from MOF and hypoxic neuro-logical injury	Decreased EF (45–50%)	Cytokine-mediated Capillary Leak Syndrome due to treatment acute toxic effect	Death (at autopsy severe DMD cardio-myopa-thy)
Case 5–7	Strauss et al 2022 ⁴¹	6 m	SMA	scAAV-9-FL-SMNCdNA	1.1×10^{14} vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 1: ↑ Tnn and Ck-MB	...	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 8–36	Gowda et al 2023 ⁴⁸	6–10 m	SMA	scAAV-9-FL-SMNCdNA	1.1×10^{14} vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 2: ↑ Tnn	Normal	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 37–54	Servais et al 2024 ⁵¹	4–14 m	SMA	scAAV-9-FL-SMNCdNA	1.1×10^{14} vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 2: ↑ Tnn	-	Probable immune response of to the vecto caspid	Re-solved during follow-up

(Continued)

Table 2. Continued

Pa-tients	Study	Age	Dis-ease	AAV delivery vector	Dose	Prede-livery immu-nosup-pression	Post-therapy immu-nosup-pression	Preex-isting condi-tions	Clinical Course	Echocar-diogram	Explanation	Out-come
Case 55–56	Weiβ et al 2022 ⁵⁰	12–16 m	SMA	scAAV-9-FL-SMNCdNA	1.1 × 10 ¹⁴ vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 1: ↑ Tnn	focal wall motion abnormalities decreased EF	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 57	Bitetti et al 2023 ⁵¹	12–16 m	SMA	scAAV-9-FL-SMNCdNA	1.1 × 10 ¹⁴ vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 2: ↑ Tnn	Normal	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 58–67	Tokaty et al 2023 ⁵⁰	6 m -18 m	SMA	scAAV-9-FL-SMNCdNA	1.1 × 10 ¹⁴ vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 2: ↑ Tnn	...	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 68–69	Stettner et al 2023 ⁵⁸	6 m	SMA	scAAV-9-FL-SMNCdNA	1.1 × 10 ¹⁴ vg/kg	Cortico-steroids (p.o.)	Cortico-steroids (p.o.)	...	Week 2: ↑ Tnn	...	Probable immune response of to the vecto caspid	Re-solved during follow-up
Case 70–71	Shieh et al 2023 ⁶⁴		X-Lynk Myotub Myop	AAV-8	3.5 × 10 ¹⁴ vg/kg	Cortico-steroids (p.o.)	Cortico-steroids + Sirolimus + Myco-phenolate Mofetil	...	Within week 3: ↑ Tnn	focal wall motion abnormalities decreased EF	Probable immune response of to the vecto caspid	Re-solved during follow-up

AAV indicates adeno-associated virus; ALT, alanine transaminase; ARDS, acute respiratory distress syndrome; AST, aspartate transaminase; DMD, Duchenne muscular dystrophy; ECMO, extracorporeal membrane oxygenation; EF, ejection fraction; LDH, lactate dehydrogenase; MOF, multi organ failure; Peric Effusion, pericardial effusion; PLT, platelet; PVC, premature ventricular contraction; and UNL, upper normal limit.

to the vector rather than a delayed transgene-specific reaction, and this event likewise improved under the documented steroid regimen (Table 2).³¹

One patient (Case 4),³³ a 27-year-old male with DMD, died after treatment with recombinant AAV serotype 9 using a dead *Staphylococcus aureus* Cas9 at high dose. The patient had a restrictive pulmonary defect, severely reduced muscle mass and mild DMD cardiomyopathy and experienced a cytokine-mediated capillary leak syndrome with cardiac dysfunction, thought to be related to an acute effect of the gene therapy. Concomitant myocarditis was possible but was likely not the primary cause of death (Table 2).³¹

Factors Associated With Occurrence of Immune-Mediated Myocardial Injury and Myocarditis

All patients received a dose >10¹³ vg/kg and all had systemic intravenous infusion of the viral vector (Tables 1 and 2). Events occurred only in patients with muscular and cardiac diseases, such as DMD, spinal muscular atrophy and X-linked myotubular myopathy and in diseases with X-linked or autosomal recessive inheritance (Figure 3). Vectors associated with the

occurrence of myocardial injury and myocarditis were recombinant AAVs (AAV serotype rh74,²⁸ recombinant AAV serotype 9 with dSaCas9,³² self-complementary AAV serotype 9 FL-survival motor neuron cDNA,^{42,49–52,59,60} 7.3 [95% CI, 5.2%–11.4%]; *P*=27.5%) and AAV serotype 8 (7.1 [95% CI, 3.8%–13.4%]; *P*=48.6%).^{26,65} AAV-1,2, 2/8, 5, 6, and 9 were not related to the occurrence of myocardial injury/myocarditis (Figure 4). No cases of myocardial injury/myocarditis were reported in patients treated with AAV gene therapy for primary cardiac diseases.

Global Pharmacovigilance Analysis

A total of 2134 and 2197 AEs have been reported in the web-based public interface to VigiBase (VigiAccess) and FAERS databases, respectively, for 5 commercially available gene-replacement drugs (Table 3). Immune-mediated AEs were 262/2134 (12%) and 448/2197 (20%). Myocarditis occurred in relation to delandistrogene moxeparvovec (1/16 [6%] in VigiAccess, 11/29 [38%] in FAERS) and onasemnogene abeparvovec (14/217 [6%] in VigiAccess, 118/395 [30%] in FAERS), although for the latter it is not known whether a baseline increase of the troponin was present.

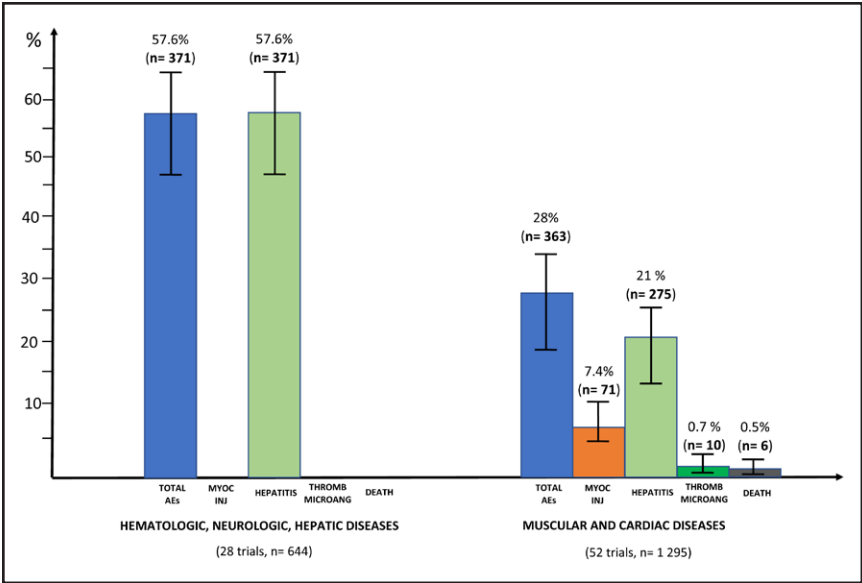


Figure 3. Prevalence of adverse events based on the type of treated disease.
AE indicates adverse event; and THROM MICROANG, thrombotic microangiopathy.

DISCUSSION

The broad clinical adoption of gene replacement therapy has been challenged by concerns over immune activation, specifically of TMA and hepatotoxicity.^{5–7} To date, the understanding of these complications has been limited and derived exclusively from individual trial reports, small case series, or company communications.⁹ We report here a comprehensive pooled analysis of the incidence and clinical significance of immune-mediated AEs after gene replacement therapy, with a specific focus on myocardial injury and myocarditis. We analyzed 80 studies encompassing 1939 patients treated

over a total of 2122 patient-years. Overall, one-third of patients experienced at least 1 immune-mediated AE. Hepatotoxicity was the most prevalent complication, followed by myocarditis and myocardial injury in the assembled data set with 71 patient-level events captured in the source reports (Table 2). However, of these, 31 cases lacked any complementary cardiac investigations or even a clinical description in the source reports.^{40,50,57,59} This paucity of objective tissue- or imaging-level adjudication severely limits mechanistic inference and complicates therapeutic decision-making.¹⁵ This alarming gap in objective adjudication may interfere with a direct patient safety concern. Without rapid,

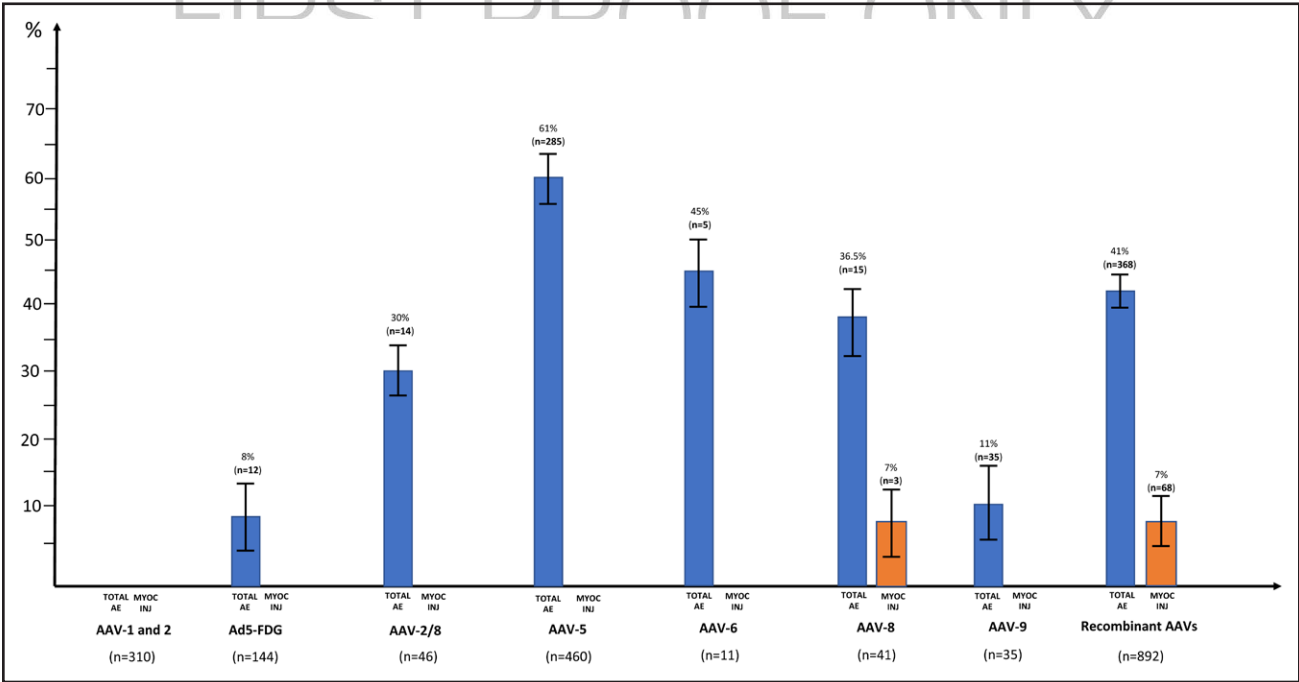


Figure 4. Pooled incidence of adverse events stratified for type of adeno-associated virus (AAV) vector.
TM indicates thrombotic microangiopathy.

Table 3. Real World Immune Adverse Events as Reported in VigiAccess and FAERS Database in Commercially Available Gene Replacement Treatments

Drug	Vector	Disease	Patients treated	Overall reported AEs		Total immune-mediated AEs		Myocarditis	
				VigiAccess	FAERS	VigiAccess	FAERS	VigiAccess	FAERS
Delandistrogene moxeparvovec	rAAVeh74	DMD	≈800	101	122	16/101 (16%)	29/122 (24%)	1/16 (6%)	11/29 (38%)
Onasemnogene abeparvovec	AAV-9	SMA	≈4000	1969	2012	217/1969 (11%)	395/2012 (20%)	14/217 (6%)	118/395 (30%)
Etranacogene dezaparvovec	AAV-5	Hem B	NA	23	22	18/23 (78%)	5/22 (23%)	0	0
Valoctocogene roxaparvovec	AAV-5	Hem A	NA	20	34	11/20 (55%)	19/34 (56%)	0	0
Eladocogene exuparvovec	AAV2-hRPE65v2	AADC deficiency	NA	21	6	0	0	0	0

AAV indicates adeno-associated virus; AE, adverse event; and FAERS, FDA Adverse Event Reporting System.

standardized diagnostic pathways clinicians may either undertreat fulminant inflammatory disease or over-treat benign cardiac biomarker elevation.¹⁰² For patient safety, we think that an unexplained rise in myocardial biomarkers after AAV exposure must trigger a mandatory, time-bound diagnostic cascade, as suggested by current guidelines.¹⁶ Implementing such protocolized safeguards is ethically imperative: it protects participants in ongoing studies, improves the accuracy of case classification, and generates the high-quality evidence needed to reduce preventable cardiac harm in future gene therapy programs.

Timing and Clinical Significance of Immune-Mediated AEs

While AAV-mediated immunotoxicity is relatively common, most events cluster in the early post-treatment period and are transient or mild in severity.¹⁰³ In our pooled data set, myocarditis/myocardial injury was uncommon but clinically meaningful, occurring in 6.2% of patients and clustering temporally in the early postinfusion window, with most events presented within 1 to 3 weeks and peaked around week 2 (Figure 2). The clinical and temporal pattern in these cases supports 2 pathophysiologic phenotypes with distinct potential implications for surveillance and management: (1) an immediate/early syndrome (hours–days) dominated by systemic inflammatory features and secondary myocardial injury, complement activation, and endothelial injury that can be associated with TMA; and (2) a subacute, transgene-directed myocarditis (typically week 1–3), more consistent with common viral myocarditis.⁴ No events occurred after the first 3 weeks after injection. These observations argue for concentrated early cardiac monitoring to map clinical phenotypes and to guide targeted prophylaxis or therapy. Clinically, the great majority were self-limited, with ≈98% of myocarditis/myocardial injury episodes resolving during study follow-up. Eight patients developed cardiac dysfunction and

wall motion abnormalities on echocardiogram. All the cases were treated with pre and postinjection corticosteroids, and 2 cases needed also second and third-line immunosuppression (eculizumab, tocilizumab, sirolimus and mycophenolate mofetil).

Factors Associated With Immune-Mediated Myocardial Injury and Myocarditis

All myocarditis/myocardial-injury events in our assembled data set occurred after systemic (intravenous) AAV delivery at doses $>1 \times 10^{13}$ vg/kg. They were reported exclusively in programmes treating primary muscular or cardiac disorders (predominantly DMD, spinal muscular atrophy, and X-linked tubular myopathy; Tables 2; Figure 3). Such conditions, which are X-linked or autosomal-recessive are commonly associated with absent or markedly altered native protein (no cross-reactive immunologic material). Whether the lack of prior antigen exposure can increase the risk of transgene-directed immune responses and be responsible for the increased number of acute myocardial injury/myocarditis is interesting but needs experimental confirmation. In pooled analyses, the signal clustered with certain recombinant platforms represented in the data set (notably AAV serotype rh74, recombinant AAV serotype 9 constructs carrying dSaCas9 or self-complementary AAV serotype 9 full-length survival motor neuron cassettes) and with AAV serotype 8 (pooled point estimates ≈7%), whereas AAV-1/2/2-8/5/6/9 did not show an association in these cohorts. Interpreting these serotype–signal patterns requires caution: capsid choice, cassette design, administered dose and indication are strongly colinear in current programs, so apparent capsid effects may largely reflect high systemic dose and indication-specific vulnerability rather than an intrinsic cardiotoxicity of a particular serotype. Mechanistically, the concentration of events after high systemic exposure and in multisystem disease cohorts argues for conservative, stepwise dose escalation in systemic programs, preferential consideration of delivery routes

that limit systemic exposure for myocardial targets, and mandatory early cardiac surveillance in high-risk indications. Still, there is a critical need for mechanistic studies, both in preclinical models and in early-phase trials, to systematically evaluate cytokine profiles, complement activation markers, endothelial injury assays, and histological characterization, alongside pharmacokinetic and pharmacodynamic parameters.^{4,12,104} A systematic evaluation at the study level of the potential effect of combined regimens was not feasible, due to the small number of studies (n=10), limited sample size, and heterogeneity of timing/dosing. Moreover, the analysis could have been potentially confounded by indication and dose (many programs employing combined regimens were also high-dose systemic programs). Therefore, future prospective investigations are much needed to test whether tailored immunosuppressive regimens (eg, complement inhibitors, transient B-cell depletion) can interrupt these pathways without blunting transgene expression or excessively immunosuppressing the patient.

Pharmacovigilance Analysis

Pharmacovigilance data reinforce the myocarditis signal observed in clinical studies but also highlight important uncertainty in real-world reporting. In 2 large AE repositories myocarditis was conspicuously reported after delandistrogene moxeparvovec (1/16, 6% in VigAccess; 11/29, 38% in FAERS) and after onasemnogene abeparvovec (14/217, 6% in VigAccess; 118/395, 30% in FAERS), whereas other approved AAV products had far fewer myocarditis entries (Table 3). These proportions, particularly the much higher percentages seen in FAERS, must be interpreted cautiously: absolute case counts are small in some denominators, spontaneous reporting systems differ in capture methods and coding, and case adjudication can be inconsistent across databases. Moreover, in these databases, an inherent bias of passive surveillance, potentially underestimating the prevalence of AEs should be acknowledged. Despite these limitations, the concordance between pharmacovigilance signals and our pooled trial data strengthens the concern that myocarditis/myocardial injury is a real, program-relevant complication of some systemic AAV programmes.

Finally, we report also the need for systematic reporting of AE with rigorous scientific studies. At least 5 cases of TMA, 1 of myocarditis and 3 deaths could have not been included in the present analysis because described in company statements.^{105–108} The impact on the pooled incidence would have not been substantial, but the absence of transparent and scientifically rigorous reporting are missed opportunities to improve the understanding of the underlying pathophysiologic mechanisms and their prevention.¹⁰⁹

Proposed Practical Clinical Approach to Suspected Acute Myocardial Injury or Myocarditis Following AAV Gene Therapy

Based on these findings, we propose a pragmatic and clinically oriented approach to suspected acute myocardial injury or myocarditis in the context of AAV gene therapy (Figure 5).

- Baseline assessment (preinfusion): history/physical, baseline high-sensitivity cTn, 12-lead electrocardiogram (ECG), and review of comorbidities/medications. Consider cardiology consultation for patients with pre-existing cardiac disease.¹⁵
- Surveillance (postinfusion): daily vital signs and symptom assessment for 48 h to 72 hours; high-sensitivity cTn and ECG daily for the first 7 days, then twice weekly through day 28, then weekly to day 56 for high-risk protocols. Shorter monitoring may be reasonable for low-risk programmes that use low systemic doses/delivery routes minimizing capsid exposure.
- Escalation triggers: new/worsening cardiac symptoms, high-sensitivity cTn rise $>99^{\text{th}}$ percentile and >1.5 baseline (or absolute rise above threshold), new ECG changes, or new echocardiographic wall-motion abnormalities or left ventricular ejection fraction drop $>5\%$ to 10% from baseline.¹⁶
- Diagnostic cascade on trigger: immediate hospital evaluation, repeat ECG and high-sensitivity cTn, urgent transthoracic echo, and, when available/feasible expedited cardiovascular magnetic resonance (T2/late gadolinium) to help identify inflammation/edema.¹⁵ Endomyocardial biopsy is reserved for cases where noninvasive testing is inconclusive or fulminant disease is present and biopsy results would change therapy (eg, identify viral inclusion bodies, or evidence of complement-mediated microangiopathy).¹¹⁰
- Initial treatment: begin systemic corticosteroids for suspected immune-mediated myocarditis (eg, prednisone 1 mg/kg per day or equivalent; for severe/fulminant presentations consider intravenous methylprednisolone pulses with cardiology/intensive care unit oversight).^{15,16} For suspected complement-driven TMA/capillary leak or severe multiorgan involvement, early multidisciplinary input and consideration of complement inhibition (eculizumab), therapeutic plasma exchange, or targeted biologic therapy (eg, anti-IL6) should be considered in consultation with hematology/immunology. For suspected transgene-directed cytotoxic T-cell responses, if available and appropriate, transient B-cell depletion (rituximab) and other targeted approaches may be considered in clinical trial contexts.¹¹¹
- Documentation and reporting: protocolized case reporting, harmonized event adjudication, and banked biospecimens (blood for cytokines, complement markers, anticapsid and antitransgene antibodies, and,

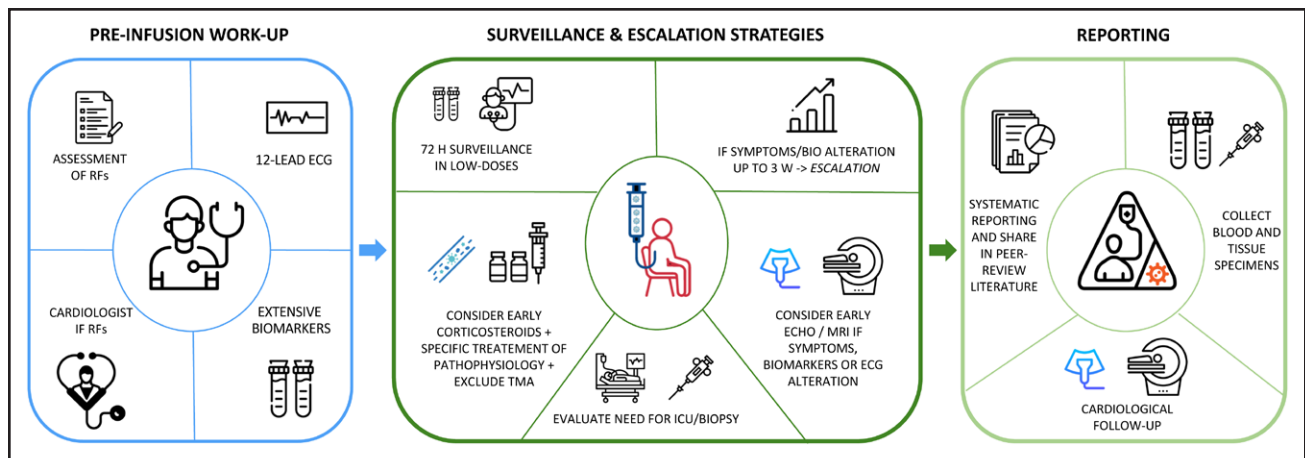


Figure 5. Proposed pragmatic and clinical oriented approach to suspected acute myocardial injury or myocarditis in the context of adeno-associated virus (AAV)-gene therapy.

H indicates hour; ICU, intensive care unit; MRI, magnetic resonance imaging; RF, risk factor; TMA, thrombotic microangiopathy; and W, week.

when available, myocardial tissue) to enable mechanistic study.¹⁰⁹

Limitations

This study has important limitations rooted in the underlying evidence base. Many included reports were small and carried moderate risk of bias, and the trials and series differed substantially in eligibility criteria and surveillance intensity—features that produce marked between-study heterogeneity. To address this, we used random-effects models with the Hartung-Knapp-Sidik-Jonkman method and performed meta-regression to probe sources of heterogeneity, but residual confounding undoubtedly remains. Small-study effects and publication bias may have inflated pooled incidence estimates, particularly for overall AE counts and hepatotoxicity. Our synthesis relied on published aggregate data, which prevents robust adjustment for patient-level confounders (eg, baseline disease severity or concomitant therapies) and limits causal inference. Differential follow-up durations and variable AE surveillance windows across studies may have caused under-ascertainment of late or sub-clinical immunotoxicity. Likewise, the pharmacovigilance analyses are constrained by passive reporting and limited clinical adjudication, reducing the precision of real-world risk estimates. Sex and gender of individuals experiencing AEs were not systematically reported, for statistical coherence was not part of the analysis. Finally, while this article was in preparation 2 additional fatal events were reported to regulators: 1 death in a participant in the RP-A501 phase-2 program associated with a capillary leak syndrome in the context of a novel pretreatment immunomodulatory regimen (the US Food and Drug Administration placed the trial on clinical hold on May 23, 2025),¹⁰⁷ and 1 death after delandistrogene moxeparvovec-rokl related to acute liver failure despite

corticosteroid use.¹⁰⁸ These developments further underscore the need for cautious interpretation, ongoing post-marketing surveillance, and prospective, harmonized safety data collection.



Conclusions

Immune-mediated myocarditis/myocardial injury after systemic AAV gene replacement therapy occurred in 6.2% of treated patients, peaking at week 2 and absent after the first month. Clinically, most episodes were mild, with 94% who had no major functional consequence, while 6% developed transient left ventricular dysfunction or wall-motion abnormalities that recovered on follow-up; 1 death occurred in the context of a cytokine-mediated capillary leak syndrome with cardiac dysfunction. Myocarditis was observed exclusively after systemic (intravenous) dosing $>1 \times 10^{13}$ vg/kg, concentrated in programmes for neuromuscular disorders, and clustered with certain recombinant platforms (notably AAV serotype rh74/rAAV9 constructs and AAV serotype 8). These findings have clear operational implications: early, protocolized cardiac surveillance (troponin/ECG and prompt imaging) during the first month after dosing and pretrial stratification by dose, vector platform, and indication. Prospective, harmonized adjudication of cardiac events combined with mechanistic biomarker studies to guide tailored immunomodulatory strategies.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S6

Figures S1–S5

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